

Prototype Test Outline

Prototype Test – Weapon

Subtest	Number of Weapons	Quantity of Ammunition per Weapon				Total Rounds			
		Training Round	Counter Defilade Round	Close Quarters Battle round	Counter-sUAS round	Training Round	Counter Defilade Round	Close Quarters Battle round	Counter-sUAS round
Inspection	12	25	0	0	0	300	0	0	0
Parts Interchange	12	35	0	0	0	420	0	0	0
Zero Repeatability	3	0	0	0	0	0	0	0	0
Zero Retention	1	20	20	20	20	20	20	20	20
Recoil	1	32	32	32	32	32	32	32	32
Kinematics	1	10	10	10	10	10	10	10	10
Reliability/Durability	5	2700	100	100	100	8100	300	300	300
Firing Table - Ballistic Match	1	165	120	0	0	165	120	0	0
High Temperature									
Operational	3	730	15	70	15	2190	45	210	45
Extreme	1	25	32	15	32	25	32	15	32
Low Temperature									
Operational	3	1100	15	100	15	3300	45	300	45
Extreme	1	25	32	15	32	25	32	15	32
Ricochet	1	250	0	0	0	250	0	0	0
Attitudes	3	30	0	0	0	90	0	0	0
Noise	1	8	0	0	0	8	0	0	0
Toxic Fumes	1	32	32	32	32	32	32	32	32
Flash	1	32	32	32	32	32	32	32	32
Unlubricated	3	320	0	0	0	960	0	0	0
Thermal Shock	3	30	10	0	0	90	30	0	0
Altitude	3	30	10	0	0	90	30	0	0
Solar Radiation	3	30	10	0	0	90	30	0	0
Dust	3	50	25	0	0	150	75	0	0
Wet Mud	3	30	0	0	0	90	0	0	0
Dry Mud	3	30	0	0	0	90	0	0	0
Humidity	3	40	20	0	0	120	60	0	0
Rain	3	120	0	0	0	360	0	0	0
Fresh Water Immersion	3	30	10	0	0	90	30	0	0
Salt Fog	3	40	20	0	0	120	60	0	0
1.5-m (5ft) Drop onto Concrete	1	5	0	0	0	5	0	0	0
Chemical Compatibility	Coupons Only	0	0	0	0	0	0	0	0

NOTE: All tests will be performed in general accordance with Test Operations Procedure 03-2-045A Small Arms - Hand and Shoulder Weapons and Machine Guns. All tests, test methods, and quantities are approximations and are subject to change.

Prototype Test Outline

Prototype Test – Ammunition

Subtest	Number of Weapons	Number of EPVAT Barrels	Quantity of Ammunition per Weapon				Total Rounds			
			Training Round	Counter Defilade Round	Close Quarters Battle round	Counter UAS round	Training Round	Counter Defilade Round	Close Quarters Battle round	Counter-sUAS round
Dispersion	5	0	64	64	64	64	320	320	320	320
Sequential Rough Handling	1	0	320	320	320	320	320	320	320	320
Electronic Pressure, Velocity, and Action Time (EPVAT)	0	2	50	50	50	50	50	50	50	50
Airburst Accuracy	2	0	0	160	0	160	0	320	0	320
Airburst Confirmation (following Parts Interchange)	2	0	0	3	0	3	0	6	0	6
Fuze Arming	1	0	0	64	0	64	0	64	0	64
Fuze Safety (sensitivity, graze, neutralization, ricochet)	1	0	0	240	0	240	0	240	0	240
Point Detonation	1	0	0	16	0	16	0	16	0	16
Long-Term Storage	1	0	5	40	40	40	5	40	40	40
C-sUAS Targets (Sphere)	1	0	0	0	0	150	0	0	0	150

NOTE: All tests will be performed in general accordance with Test Operations Procedure 03-2-045A Ammunition Small Arms. All tests and quantities are approximations and are subject to change.

Prototype Test Outline

Prototype Test – Fire Control

Subtest	Number of Fire Controls
Inspection	24
LRF Accuracy	3
0.9-m (3 ft) Drop onto Dirt	3
Field of View	3
Diopter Adjustment	3
Diopter Uniformity and Retention	3
Exit pupil	3
Clear Eye Distance	3
Target Focus	3
Magnification	3
Modulation Transfer Function	3
Resolution Limit	3
Transmittance	3
Optical Coatings (Need Samples)	3
Veiling Glare	3
Passive Reticle Focal Plane	1
Passive Reticle Surface Quality	3
Elevation / Windage Adjustment Range	3
Reticle Cant	3
Display Zones	1
Dynamic Reticle	1
Dynamic Reticle Focal Plane	1
Diopter Reticle Offset	3
Eye Box	3
Visibility	3
Luminance Degradation	3
Display Burn-in	3
Response Time	3
LRF Ranging Capability	3
Beam Divergence	3
Wavelength	3
Range Accuracy	3
False Alarm	1
Adjustment Range	3
Coded Spot Tracking	3
Laser Safety	1
Visible Laser	
Visible Aiming Laser Capability	3
Beam Divergence	3
Wavelength	3
Output Power and Irradiance	3
Laser Safety	3
Laser Activation	3
Infrared Aiming Laser	
Aiming Laser Capability	3
Beam Divergence	3
Wavelength	3
Output Power and Irradiance	3
Laser Safety	1
Laser Activation	3
Lens Protection	1
Operational Torque	3
Glint Suppression	3
Size (length, height, width)	1
Surface finish	3
Color	3
NIR Reflectance	3
Transport Case (if available)	1
Mount	1
Operating Orientation	1
Captive Mounting Hardware	1
Interference	1
Ballistic Laser Protective Spectacles	1
Optical Center Line	1

Subtest	Number of Fire Controls
Ranging Capability with FWS-I	1
Objective Interface	1
Laser Filter Unit	1
Anti-Reflection Device	1
Remote Mountable	1
Interconnection	1
Communication Port / Wired Port	1
Maintenance Cable	1
Wireless Transmission Distance	3
Network Configuration	3
Network Association	3
Sensors Inclination Angle	3
Sensors	
Cant Angle	3
Temperature	3
Barometric Pressure	3
Humidity	3
Power Source	1
Reverse Polarity Protection	1
Battery Orientation Marking	1
Battery Compartment Cover	1
Battery Replacement	1
Battery Level Monitor	1
Battery Run Time (all temps)	
Ambient	3
Operational Hot	3
Extreme Hot	3
Operational Cold	3
Extreme Cold	3
Software Standby Mode Inactivity	3
Start-Up/Wake-Up Time	3
System Updates	1
Ballistic Solver Computation Time	3
Target Reference Points	1
System Accuracy - Reticle Boresight Error	3
Reticle Calibration Error	3
Parallax Error	3
Magnification Switch Adjustment Error	3
Applied Forces	3
LRF Error Boresight	3
Boresight Repeatability (Reattachment)	3
Boresight Retention (Gunfire Pyroshock)*	3
Low Pressure (High Altitude)	3
Operational Temperature	5
Storage Temperature	3
Temp Shock	3
Solar Radiation	3
Transportation and Vibration	3
Humidity	3
Salt Fog	3
Immersion	5
Transport Drop	3
Optical Quality and Cleanliness	3
Gunfire During Rain*	3
Chemical Compatibility	Coupons Only
Dust*	3
Electromagnetic Effects	2
System Reliability*	3
Detectable Emissions	3
Material	1
Workmanship	3
Marking	1
Cooperative Vulnerability Identification	1
3D Print	1

NOTE: All tests and quantities are approximations and are subject to change.

***Conducted during Weapon testing**